Louisville Weather and Business Risk Dashboard

Project Proposal

Developer: Oluwatosin Adelusi

Target Audience: Louisville-area finance analysts, retail and operations managers, SMB decision-makers

Technology Stack: Python, Open-Meteo API, Pandas, CSV Storage, Plotly Dash

Background

Weather conditions exert measurable influence on consumer behavior, business operations, and financial outcomes. This is especially true in Louisville, Kentucky, where several of the region's biggest economic engines are directly exposed to weather:

- UPS Worldport :the global air-cargo hub at Louisville Muhammad Ali International Airport is sensitive to thunderstorms, high winds, and winter weather.
- Kentucky Derby Festival, Waterfront Park events, and bourbon-trail tourism all depend on favorable outdoor conditions.
- LG&E utility load and GE Appliance Park energy usage rise sharply during temperature extremes.
- Retail districts like 4th Street Live, NuLu, and Mall St. Matthews see foot traffic fall on heavy-rain and severe-weather days.
- Riverfront real estate and Ohio River logistics are exposed to flooding and severe-storm risk.

In Spite of this information, Louisville business users do not have a consolidated, decision-oriented view of weather conditions tied to the sectors that care most. Weather data lives in one place and business planning lives in another.

This project proposes an end-to-end data engineering pipeline that ingests free weather data for Louisville from the Open-Meteo API, transforms it into business-relevant signals (such as Heating and Cooling Degree Days and severe-weather flags), and visualizes the results in an interactive dashboard. The deliverables answer the questions a finance and business-analytics audience in Louisville actually asks:

- How is current weather likely to affect downtown retail foot traffic this week?
- Is energy demand trending up or down based on temperature?
- How exposed is Worldport to severe weather over the next two weeks?

The project aligns directly with the developer's background in finance and business analytics by translating Louisville environmental data into commercial insight.

Business Problem

Louisville businesses and financial analysts often lack automated systems for monitoring how local weather conditions influence consumer activity, operational efficiency, and financial planning. Specifically:

- Weather data for Louisville is available on public forecast sites but is not packaged for business decision-making.
- Raw temperature, precipitation, and wind metrics are not directly meaningful to business users they need to be translated into financial and operational signals.

- Existing dashboards focus on day-to-day forecasting, not on the downstream business impact (retail, energy, logistics, tourism, insurance).
- There is no lightweight, free tool that gives Louisville analysts a single-page view of severe-weather and degree-day risk indicators for the metro area.

This project solves these gaps by building a focused, automated pipeline that pulls Louisville weather data, derives business-relevant risk indicators, and presents them in an interactive dashboard tailored to the local economy.

Objectives

- Create a Louisville weather and business risk dashboard
- Extract real-time and forecasted weather data for Louisville, KY from the Open-Meteo REST API.
- Transform JSON responses into clean, structured tabular datasets with consistent timestamps, units, and frequencies.
- Derive business-relevant metrics tied to Louisville's economy, including:
    - Heating Degree Days (HDD) and Cooling Degree Days (CDD) for LG&E utility-load context.
    - Severe-weather and heavy-rain flags relevant to UPS Worldport and outdoor events.
    - Rolling 7-day and 30-day temperature and precipitation averages for trend analysis.
- Store the cleaned and derived data in a SQLite relational database, with CSV backup snapshots.
- Build an interactive dashboard in Power BI with at least five high-quality, decision-oriented visualizations.
- Automate the pipeline to refresh daily, with logging and basic error handling.
- Deliver a polished demo, documentation, and project report at the end of Week 5.

Technical Approach

1. Weather Data Dashboard

- **Current Conditions Display:** Real-time temperature, humidity, precipitation, and wind speed for Louisville.
- **Visual Weather Timeline:** Interactive line and bar charts showing temperature, precipitation, and wind trends over the most recent 30 days.
- **Seasonal Context:** Comparison of current values to Louisville historical norms (e.g., is this week warmer or cooler than typical?).

2. Business Risk Indicator Engine

A rules-based engine translates raw weather variables into business-friendly signals tuned for Louisville's economy:

- **Temperature signals:** Compute HDD = max(0, 65°F daily mean temp F) and CDD = max(0, daily mean temp F 65°F) to forecast Louisville utility load.
- **Precipitation signals:** Classify days as Dry, Light Rain, Moderate Rain, or Heavy Rain and flag Heavy Rain days as elevated risk for retail foot traffic and outdoor events.
- **Wind signals:** Flag days with sustained wind above 25 mph as risk events for air-cargo operations at Worldport and outdoor tourism.

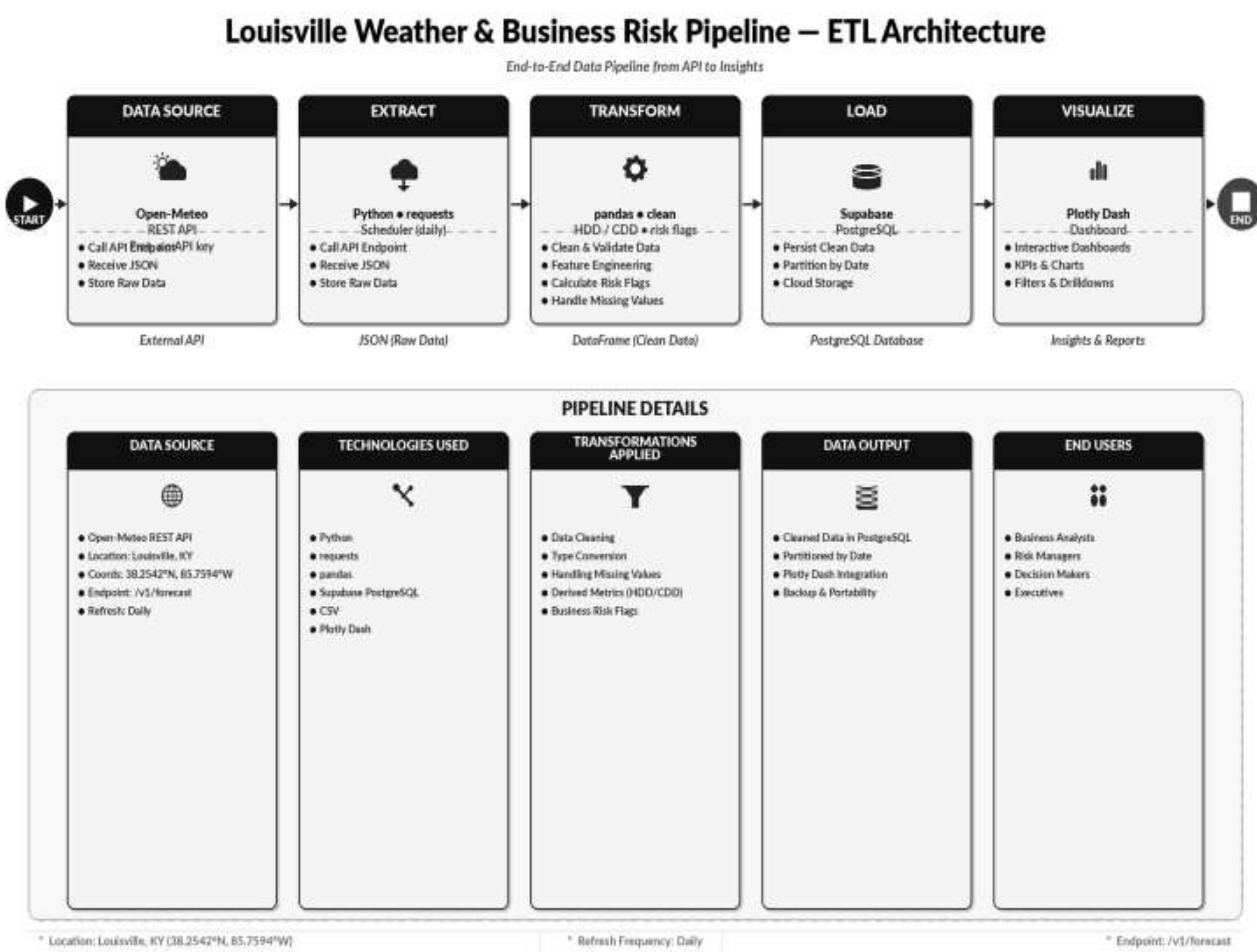
- **Severe-weather flag:** Combine extreme temperature, heavy precipitation, and high wind into a single composite risk flag for the day.
- **Louisville Weather Risk Score:** A daily composite score (0–100) summarizing overall weather-related business risk for the metro area.

3. Interactive User Interface

- **Interactive Charts:** Power BI visualizations with zoom, tooltip, and date-range selection.
- **Time Range Selectors:** Quick filters for last 7 days, last 30 days, or custom range.
- **Sector Spotlights:** Dedicated panels showing the most relevant indicator for each sector: utilities, retail, logistics, tourism, and insurance.
- **Risk Spotlight:** A dedicated panel highlighting current severe-weather flags and high-risk days in the daily forecast.

Technical Architecture

The system follows a classic Extract – Transform – Load – Visualize architecture, with each stage producing a clean handoff to the next:



1. Backend Components

- **Data Retrieval:** Open-Meteo API integration using the Python requests library to fetch real-time weather data.
- **Data Processing:** pandas for cleaning, reshaping, and computing degree days and risk flags.
- **Risk Engine:** Rule-based logic that translates weather variables into Louisville business signals.
- **Data Storage:** CSV Storage
- **Scheduling:** APScheduler triggers a daily refresh job.

2. Frontend Components

- **Visualization Layer:** Plotly Dash Desktop for building interactive dashboard.
- **UI Components:** Filters, KPI cards, multi-panel charts, sector spotlights, and a risk spotlight section.
- **Styling:** Clean, business-oriented theme with Louisville-aware color coding for risk levels.

3. Data Flow

- **API Integration:** The pipeline calls the Open-Meteo forecast endpoint at regular intervals for Louisville data.
- **Data Processing:** Raw JSON is parsed into pandas DataFrames, cleaned, aligned, and enriched with HDD/CDD and risk flags.
- **Storage:** Cleaned and derived data is saved into a csv file
- **Visualization:** The dashboard reads from CSV, renders charts and KPIs, and exposes filters for end users.
- **User Interaction:** Analysts filter by date range and sector to investigate weather trends and business exposures across Louisville.

Data Sources and APIs

Primary Data Source: Open-Meteo API

- **Endpoint:** https://api.open-meteo.com/v1/forecast
- **Authentication:** None required free and beginner-friendly REST API.
- **Location:** Louisville, KY (38.2542°N, 85.7594°W).
- **Units:** Fahrenheit (temperature), inches (precipitation), mph (wind).
- **Timezone:** America/Louisville.

Data Parameters

- **Current:** temperature2m, relative humidity 2m, apparent temperature, precipitation, weather code, cloud cover, wind speed 10m.
- **Daily:** temperature 2mmax, temperature 2m min, temperature 2m mean, precipitation sum, wind speed 10m max, weather code, sunrise, sunset.

Business Risk Indicator Logic

The risk engine translates weather variables into clear, business-relevant signals for Louisville's key sectors. The table below summarizes the mapping:

Weather Variable	Business / Financial Signal	Louisville Sector Affected
Mean temperature	Heating/Cooling Degree Days drive energy demand	LG&E utilities, GE Appliance Park
Heat / cold extremes	Reduced foot traffic, higher HVAC costs	4th Street Live, NuLu retail
Precipitation totals	In-store retail declines; logistics delays	Mall St. Matthews, UPS Worldport
Heavy rainfall flag	Outdoor event risk; flood concerns near Ohio River	Waterfront Park events, riverfront real estate
High wind speed	Air-cargo delays; event cancellations	UPS Worldport, Kentucky Derby Festival
Severe weather flag	Property damage and claims pressure	Insurance, hospitality, agriculture
UV index / sun hours	Spending shifts to outdoor dining and tourism	Bourbon Trail tourism, restaurants

Threshold Rules (illustrative)

- **Temperature:** Mean below 32°F or above 95°F is flagged as extreme.
- **Precipitation:** Daily total above 1 inch is flagged as Heavy Rain.
- **Wind:** Sustained wind above 25 mph is flagged as High Wind.
- **Composite Risk Score:** 0 = calm, 25 = mild, 50 = elevated, 75 = high, 100 = severe.

Implementation Plan

The project follows the recommended five-week structure, with each week focused on completing a specific layer of the pipeline:

Week	Tasks & Deliverables
Week 1	Implement Open Meteo API Integration and validate API response, set up Python environment and Git repo,
Week 2	Write a reusable Open-Meteo Python client for Louisville, configure daily scheduling, add error handling and retry logic, implement data

	transformation and cleaning using pandas, build business risk indicator engine
Week 3	Build the dashboard in Plotly Dash: KPI tiles, trend charts, severe-weather risk indicators, and Louisville sector spotlights.
Week 4	Test decision rules for consistency and correctness, optimize data processing pipeline performance, validate dashboards usability.
Week 5	End-to-end testing, performance tuning, README and documentation, demo video, slide deck, and final submission.

Expected Outcomes

By the end of Week 5, the project will deliver:

- **A working end-to-end ETL pipeline** that automatically pulls fresh Louisville weather data from Open-Meteo each day.
- **An interactive dashboard** in Plotly Dash with at least 5–7 decision-oriented visualizations and sector spotlights.
- **A composite Louisville Weather Risk Score** translating raw weather data into a single, business-readable daily indicator.
- **A GitHub repository** with clean code, README, requirements.txt, and reproducible setup instructions.
- **Slide deck** presenting the pipeline, dashboard, and 2–3 analytical insights surfaced from the data.

Success Metrics

Technical Metrics

- **API Reliability:** Greater than 99% successful daily API calls.
- **Page Load Time:** Dashboard initial load under 3 seconds.
- **Data Freshness:** Weather data refreshed within 24 hours of the latest available observation.

User Experience Metrics

- **Clarity:** A first-time Louisville business user can identify the current weather risk level within 30 seconds.
- **Interactivity:** All filters update charts within 1 second of selection.
- **Insight Density:** Each dashboard view answers at least one specific business question.

Business Impact Metrics

- **Decision Support:** Output supports retail, energy, logistics, tourism, and insurance use cases described in the Business Problem section.
- **Educational Value:** Demonstrates how data engineering can convert public data into local business insight.

- **Reproducibility:** Another analyst can clone and run the project end-to-end from the GitHub README.

Risk Assessment and Mitigation

Technical Risks

- **API Rate Limiting:** Open-Meteo is generous, but the pipeline will cache responses locally and use exponential backoff on errors.
- **Data Accuracy:** Validate that key fields are within plausible ranges before persisting; log anomalies for review.
- **Performance:** Pre-compute derived metrics during the Load step so the dashboard renders cleanly.

Project Risks

- **Scope Creep:** Indicator set is locked in Week 1; additional indicators are only added if Weeks 2–3 finish ahead of schedule.
- **Timeline Slippage:** Each weekly milestone includes a clearly defined minimum deliverable so the project remains complete even if optional features are dropped.

Budget and Resources

Development Resources

- Libraries: Python Pandas, Requests (all free/open-source)
- Visualization: Plotly Dash dashboards (free)
- Data: Open-Meteo API (free, no API key required)
- Data Storage : CSV storage

Time Investment

- Development: 5 weeks of intensive development
- Testing: Integrated throughout each week
- Documentation: Ongoing throughout development
- Maintenance: 2-4 hours per month for updates and bug fixes

Conclusion

This Louisville Weather and Business Risk Dashboard represent a powerful opportunity to bridge the gap between environmental data engineering and corporate financial analysis. By translating raw meteorological variables into localized operational signals such as utility-driven degree days and logistics risk flags the project demonstrates how open-source data pipelines can convert public resources into high-value commercial intelligence for the Louisville economy.

Ultimately, this application serves as a blueprint for decision-oriented analytics. It provides retail, logistics, and energy professionals with a consolidated, single pane view to proactively mitigate weather exposure, while demonstrating a rigorous, end-to-end ETL deployment tailored to the region's major economic drivers.

Next Steps

- **Stakeholder Review:** Present this proposal to project sponsors and academic stakeholders for architectural approval.
- **Technical Feasibility Assessment:** Verify the historical limits of the Open-Meteo API data stream and test local Windows/Plotly Dash desktop data gateway connections.
- **Prototype Development:** Create a Minimum Viable Product (MVP) script to successfully parse a daily forecast payload into a test CSV.
- **User Research:** Gather feedback from local business analysts to fine-tune the thresholds for the composite Louisville Weather Risk Score.
- **Project Kickoff:** Begin Phase 1 development in Week 1 with finalized repository structures and environment configurations.